

A. Proof of Lemma 2

Lemma 2. For a positive definite matrix $\mathbf{A}$ and a positive semi-definite matrix $\mathbf{B}$, we have

$$\begin{aligned} & \text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \right) \\ & \leq \text{tr} \left(\mathbf{X}^H \mathbf{B} \mathbf{X} (\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \mathbf{A} (\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right) \\ & \quad - 2\Re \left(\text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right) \right) + \text{tr}(\mathbf{A}) \\ & \quad + \text{tr} \left((\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \mathbf{A} (\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X})^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right). \end{aligned}$$

Proof. Considering the eigenvalue decomposition of the positive semi-definite matrix

$$\mathbf{B} = [\mathbf{P}^H \quad \mathbf{Q}^H] \begin{bmatrix} \mathbf{\Lambda} & \\ & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{P} \\ \mathbf{Q} \end{bmatrix} = \mathbf{P}^H \mathbf{\Lambda} \mathbf{P},$$

where $\mathbf{\Lambda} \in \mathbb{C}^{r \times r}$ and r is the rank of matrix $\mathbf{B}$. Denote $\mathbf{Y} = \mathbf{P} \mathbf{X}$, according to matrix inversion lemma

$$\text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \right) = \text{tr}(\mathbf{A}) - \text{tr} \left(\mathbf{A}^{1/2} \mathbf{Y}^H (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H)^{-1} \mathbf{Y} \mathbf{A}^{1/2} \right).$$

Further majorization can be constructed by using the following lemma.

Lemma 4. For matrix $\mathbf{A}$ and nonsingular symmetric matrix $\mathbf{B}$, we have

$$\text{tr}(\mathbf{A}^H \mathbf{B}^{-1} \mathbf{A}) \geq 2\Re[\text{tr}(\mathbf{A}^H \mathbf{B}_k^{-1} \mathbf{A})] - \text{tr}(\mathbf{B}_k^{-1} \mathbf{A} \mathbf{A}^H \mathbf{B}_k^{-1} \mathbf{B}).$$

According to lemma 4

$$\begin{aligned} \text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \right) &= \text{tr}(\mathbf{A}) - \text{tr} \left(\mathbf{A}^{1/2} \mathbf{Y}^H (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H)^{-1} \mathbf{Y} \mathbf{A}^{1/2} \right) \\ &\leq \text{tr}(\mathbf{A}) - 2\Re \left(\text{tr} \left(\mathbf{C} \mathbf{Y} \mathbf{A}^{1/2} \right) \right) + \text{tr}(\mathbf{C}^H \mathbf{C} (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H)), \end{aligned}$$

where $\mathbf{C} = \mathbf{A}^{1/2} \mathbf{Y}^H (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H)^{-1}$. Rewrite matrix $\mathbf{C}$ by matrix inversion lemma

$$\begin{aligned} \mathbf{C} &= \mathbf{A}^{1/2} \mathbf{Y}^H (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H)^{-1} \\ &= \mathbf{A}^{1/2} \mathbf{Y}^H \left(\mathbf{\Lambda} - \mathbf{\Lambda} \mathbf{Y} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y}) \mathbf{Y}^H \mathbf{\Lambda} \right) \\ &= \mathbf{A}^{1/2} \left(\mathbf{I} - \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \right) \mathbf{Y}^H \mathbf{\Lambda} \\ &= \mathbf{A}^{1/2} \left(\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} \right)^{-1} \mathbf{Y}^H \mathbf{\Lambda}. \end{aligned}$$

Since matrix $\mathbf{C} = \mathbf{A}^{1/2} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda}$, we have

$$\begin{aligned} & \text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \right) \\ & \leq \text{tr}(\mathbf{A}) - 2\Re \left(\text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} \right) \right) \\ & \quad + \text{tr} \left(\mathbf{\Lambda} \mathbf{Y} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda} (\mathbf{\Lambda}^{-1} + \mathbf{Y} \mathbf{Y}^H) \right) \\ & = \text{tr} \left(\mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} \right) - 2\Re \left(\text{tr} \left(\mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} \right) \right) \\ & \quad + \text{tr}(\mathbf{A}) + \text{tr} \left((\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{A} (\mathbf{I} + \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y})^{-1} \mathbf{Y}^H \mathbf{\Lambda} \mathbf{Y} \right). \end{aligned}$$

Bring $\mathbf{Y} = \mathbf{P}\mathbf{X}$ and $\mathbf{P}^H\mathbf{\Lambda}\mathbf{P} = \mathbf{B}$ into the above inequality, we finally have

$$\begin{aligned}
& \text{tr} \left(\mathbf{A} \left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \right) \\
& \leq \text{tr} \left(\mathbf{X}^H \mathbf{B} \mathbf{X} \left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \mathbf{A} \left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right) \\
& \quad - 2\Re \left(\text{tr} \left(\mathbf{A} \left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right) \right) + \text{tr} (\mathbf{A}) \\
& \quad + \text{tr} \left(\left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \mathbf{A} \left(\mathbf{I} + \mathbf{X}^H \mathbf{B} \mathbf{X} \right)^{-1} \mathbf{X}^H \mathbf{B} \mathbf{X} \right).
\end{aligned}$$

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